Generated Transformation Rules Report

1. Overview

Compiled **12 rules** from the supplied R2RML mappings:

Rule category	Count
CTR	3
OTR	1
Local DTR	6
Path DTR	2
Total	**12**

The `owl:sameAs` mapping is classified as an **OTR candidate**, but the supplied metadata does not establish a target pivot relation or target CTR for its external IRI. Its rule is therefore given as a provisional operational formula, **not as a verified OTR conforming to the framework**. The annotation path is also incompletely validated because the supplied metadata does not name all required foreign-key constraints.

The subject URI predicates used below are:

- `URI_artist(a,s) ≡ hasURI("http://musicbrainz.org/artist/{gid}#_", ⟨a.gid⟩, s)`
- `URI_medium(m,s) ≡ hasURI("http://musicbrainz.org/record/{id}#_", ⟨m.id⟩, s)`
- `URI_track(t,s) ≡ hasURI("http://musicbrainz.org/track/{id}#_", ⟨t.id⟩, s)`

The distinct-tuple identity guarantee for these predicates is **conditional**: the supplied metadata does not establish that `artist.gid`, `medium.id`, or `track.id` is unique. In the specifications, `RDFLiteral(x,A,R,v)` denotes the literal representation of value `x` from attribute `A` of relation `R`; `nonNull(x)` denotes that `x` is not null.

2. Compiled Transformation Rules

2.1 Artist class

- **TriplesMap Identifier**: `lb:Artist`
- **Rule Identifier**: `\psi_artist_01`
- **Rule Type**: CTR
- **Pivot Relation**: `artist`
- **Formal TR Specification**:

  $$[\psi_{\text{artist_01}}: \quad \texttt{mo:MusicArtist}(s) \leftarrow \texttt{artist}(a),\ URI_{\texttt{artist}}(a,s)]$$

  Subject identity is conditional on `artist.gid` being unique and template instantiation being collision-free.
- **Relational Path**: None; this is a local pivot rule.

2.2 Artist MusicBrainz GUID

- **TriplesMap Identifier**: `lb:Artist`
- **Rule Identifier**: `\psi_artist_02`
- **Rule Type**: Local DTR
- **Pivot Relation**: `artist`
- **Formal TR Specification**:

  $[\psi_{\text{artist_02}}: \quad \texttt{mo:musicbrainz_guid}(s,v) \leftarrow \texttt{artist}(a),\ URI_{\texttt{artist}}(a,s), \ nonNull(a.gid), \ RDFLiteral(a.gid,\texttt{"gid"},\texttt{artist},v)]$

  The mapped literal datatype is `xsd:string`. Subject identity is conditional on `artist.gid` being unique.

- **Relational Path**: None; local DTR attribute `artist.gid`.

2.3 Artist name

- **TriplesMap Identifier**: `lb:Artist`
- **Rule Identifier**: `\psi_artist_03`
- **Rule Type**: Local DTR
- **Pivot Relation**: `artist`
- **Formal TR Specification**:

  $[\psi_{\text{artist_03}}: \quad \texttt{foaf:name}(s,v) \leftarrow \texttt{artist}(a),\ URI_{\texttt{artist}}(a,s), \ nonNull(a.name), \ RDFLiteral(a.name,\texttt{"name"},\texttt{artist},v)]$

  The mapped literal datatype is `xsd:string`. Subject identity is conditional on `artist.gid` being unique.

- **Relational Path**: None; local DTR attribute `artist.name`.

2.4 Artist sort label

- **TriplesMap Identifier**: `lb:Artist`
- **Rule Identifier**: `\psi_artist_04`
- **Rule Type**: Local DTR
- **Pivot Relation**: `artist`
- **Formal TR Specification**:

  $[\psi_{\text{artist_04}}: \quad \texttt{ov:sortLabel}(s,v) \leftarrow \texttt{artist}(a),\ URI_{\texttt{artist}}(a,s), \ nonNull(a.sort_name), \ RDFLiteral(a.sort_name,\texttt{"sort_name"},\texttt{artist},v)]$

  The mapped literal datatype is `xsd:string`. Subject identity is conditional on `artist.gid` being unique.

- **Relational Path**: None; local DTR attribute `artist.sort_name` .

2.5 Artist external DBpedia link

- **TriplesMap Identifier**: `lb:artist_dbpedia`
- **Rule Identifier**: `\psi_artist_14`
- **Rule Type**: OTR — **candidate; not verified as a conforming OTR**
- **Pivot Relation**: `artist` for the subject. No target pivot relation is established for the external IRI.
- **Formal TR Specification**:

  The framework's OTR pattern requires a target pivot tuple and a target CTR that generates the object resource. Neither is supplied for the transformed external IRI, so a verified, framework-conforming OTR cannot be formed. The following is the provisional operational rule represented by the mapping:

  $$
  \begin{aligned} \psi_{\text{artist_14}}^{\text{candidate}}: \quad \texttt{owl:sameAs}(s,o) \leftarrow & \texttt{artist}(a), \\ & URI_{\texttt{artist}}(a,s), \\ & \texttt{l_artist_url}(x), \\ & \texttt{link}(l), \\ & \texttt{link_type}(lt), \\ & \texttt{url}(u), \\ & \delta_{\text{dbpedia}}(lt,u), \\ & \texttt{nonNull}(u.url), \\ & u'=\operatorname{REPLACE}(\operatorname{REPLACE}(u.url, \texttt{'wikipedia.org/wiki'}, \texttt{'dbpedia.org/resource'}), \texttt{'http://en.'}, \texttt{'http://'}), \\ & \operatorname{RDFIRI}(u',o), \\ & \phi_{\text{dbpedia}}(a,x,l,lt,u) \end{aligned}
  $$

  Here, δ_{dbpedia} is the mapping selection condition:

  $$
  lt.gid=\texttt{'29651736-fa6d-48e4-aadc-a557c6add1cb'} \quad\land\quad u.url \texttt{SIMILAR TO} \texttt{'http://(de|el|en|es|ko|pl|pt).wikipedia.org/wiki/%'}
  $$

  The `owl:sameAs` object is intended to be an IRI, but the supplied metadata does not establish the source URL's schema type or prove that every transformed value is a valid IRI. This formula captures the mapping's apparent intent; it does **not** establish the target-pivot correspondence required by a conforming OTR.

- **Relational Path**: The intended traversal is:

  $$
  \phi_{\text{dbpedia}}: \quad \texttt{artist.id} \rightarrow \texttt{l_artist_url.entity0} \rightarrow \texttt{l_artist_url.link} \rightarrow \texttt{link} \rightarrow \texttt{link_type} \quad\text{and}\quad \texttt{l_artist_url.entity1} \rightarrow \texttt{url.id}
  $$

  This path is placed as the final body term. Only `l_artist_url_fk_entity0` is named in the supplied metadata; the complete path is therefore **not schema-validated**. The SQL join `link_type = link_type.id` is ambiguous or erroneous as written and must be corrected or clarified before this rule can be validated.

2.6 Artist annotation comment

- **TriplesMap Identifier**: `lb:artist_annotation`
- **Rule Identifier**: `\psi_artist_13`
- **Rule Type**: Path DTR
- **Pivot Relation**: `artist`

- **Formal TR Specification**:

  [\begin{aligned} \psi_{\text{artist_13}}: \quad \texttt{rdfs:comment}(s,v) \leftarrow;& \texttt{artist}(a),\ URI_{\texttt{artist}}(a,s),\ &\texttt{annotation}(n),
  \texttt{nonNull}(n.text),
  \texttt{RDFLiteral}(n.text,\texttt{"text"},\texttt{annotation},v),\ &\phi_{\text{annotation}}(a,n) \end{aligned}]

  The mapped literal datatype is `xsd:string`. Subject identity is conditional on `artist.gid` being unique.

- **Relational Path**: The intended artist-to-annotation traversal is:

  [\phi_{\text{annotation}}: \quad \texttt{artist.id} \rightarrow \texttt{artist_annotation.artist} \rightarrow \texttt{artist_annotation.annotation} \rightarrow \texttt{annotation.id}]

  This path is the final body term. The supplied metadata names `artist_annotation_fk_artist`, but does not name the annotation-side FK; consequently, the complete path is not schema-validated. The `text` column's schema type and nullability are also not supplied.

2.7 Artist gender

- **TriplesMap Identifier**: `lb:artist_gender`
- **Rule Identifier**: `\psi_artist_07`
- **Rule Type**: Path DTR
- **Pivot Relation**: `artist`
- **Formal TR Specification**:

  [\begin{aligned} \psi_{\text{artist_07}}: \quad \texttt{foaf:gender}(s,v) \leftarrow;& \texttt{artist}(a),\ URI_{\texttt{artist}}(a,s),\ &\texttt{gender}(g),
  \texttt{nonNull}(g.name),\ &\texttt{RDFLiteral}(g.name,\texttt{"name"},\texttt{gender},v_0),\ &\operatorname{LOWER}([v_0],v),
  F_{\texttt{artist_fk_gender}}(a,g) \end{aligned}]

  The mapping's value transformation is `LOWER(gender.name)` and the declared RDF datatype is `xsd:string`. Subject identity is conditional on `artist.gid` being unique.

- **Relational Path**: The final body term is the schema-validated FK traversal:

  [F_{\texttt{artist_fk_gender}}(a,g): \quad \texttt{artist.gender} \rightarrow \texttt{gender.id}]

  The supplied metadata does not state the type or nullability of `gender.name`.

2.8 Medium class

- **TriplesMap Identifier**: `lb:Medium`
- **Rule Identifier**: `\psi_medium_01`
- **Rule Type**: CTR
- **Pivot Relation**: `medium`

- **Formal TR Specification**:

  $$[\psi_{\text{medium_01}}: \quad \texttt{mo:Record}(s) \leftarrow \texttt{medium}(m),\ URI_{\texttt{medium}}(m,s)]$$

  Subject identity is conditional on `medium.id` being unique and template instantiation being collision-free. Confirm that representing medium rows as `mo:Record` in the `/record/` namespace is intentional.

- **Relational Path**: None; this is a local pivot rule.

2.9 Medium title

- **TriplesMap Identifier**: `lb:Medium`
- **Rule Identifier**: `\psi_medium_05`
- **Rule Type**: Local DTR
- **Pivot Relation**: `medium`
- **Formal TR Specification**:

  $$[\psi_{\text{medium_05}}: \quad \texttt{dc:title}(s,v) \leftarrow \texttt{medium}(m),\ URI_{\texttt{medium}}(m,s), \ nonNull(m.name), \ RDFLiteral(m.name,\texttt{"name"},\texttt{medium},v)]$$

  The mapped literal datatype is `xsd:string`. Subject identity is conditional on `medium.id` being unique.

- **Relational Path**: None; local DTR attribute `medium.name`.

2.10 Medium track count

- **TriplesMap Identifier**: `lb:Medium`
- **Rule Identifier**: `\psi_medium_02`
- **Rule Type**: Local DTR
- **Pivot Relation**: `medium`
- **Formal TR Specification**:

  $$[\psi_{\text{medium_02}}: \quad \texttt{mo:track_count}(s,v) \leftarrow \texttt{medium}(m),\ URI_{\texttt{medium}}(m,s), \ nonNull(m.track_count), \ RDFLiteral(m.track_count,\texttt{"track_count"},\texttt{medium},v)]$$

  The mapped literal datatype is `xsd:int`. Subject identity is conditional on `medium.id` being unique.

- **Relational Path**: None; local DTR attribute `medium.track_count`.

2.11 Track class

- **TriplesMap Identifier**: `lb:Track_psi_track_01`
- **Rule Identifier**: `\psi_track_01`

- **Rule Type**: CTR
- **Pivot Relation**: `track`
- **Formal TR Specification**:

  $$ \psi_{\text{track_01}}: \quad \texttt{mo:Track}(s) \leftarrow \texttt{track}(t),\ URI_{\texttt{track}}(t,s) $$

  Subject identity is conditional on `track.id` being unique and template instantiation being collision-free.
- **Relational Path**: None; this is a local pivot rule.

2.12 Track number

- **TriplesMap Identifier**: `lb:Track_psi_track_02`
- **Rule Identifier**: `\psi_track_02`
- **Rule Type**: Local DTR
- **Pivot Relation**: `track`
- **Formal TR Specification**:

  $$ \begin{aligned} \psi_{\text{track_02}}: \quad \texttt{mo:track_number}(s,v) \leftarrow& \texttt{track}(t),\ URI_{\texttt{track}}(t,s),\ \\ &\delta_{\text{track_02}}(t),\ \texttt{nonNull}(t.position),\ \\ &\texttt{RDFLiteral}(t.position,\texttt{"position"},\texttt{track},v) \end{aligned} $$

  where:

  $$ \delta_{\text{track_02}}(t) \equiv t.position\ \texttt{IS NOT NULL} $$

  The mapped RDF datatype is `xsd:nonNegativeInteger`. The filter is apparently redundant given the supplied `INTEGER NOT NULL` metadata. The metadata does not establish that every `position` value is nonnegative, so conformance with the declared RDF datatype should be confirmed. Subject identity is conditional on `track.id` being unique.
- **Relational Path**: None; local DTR attribute `track.position`.